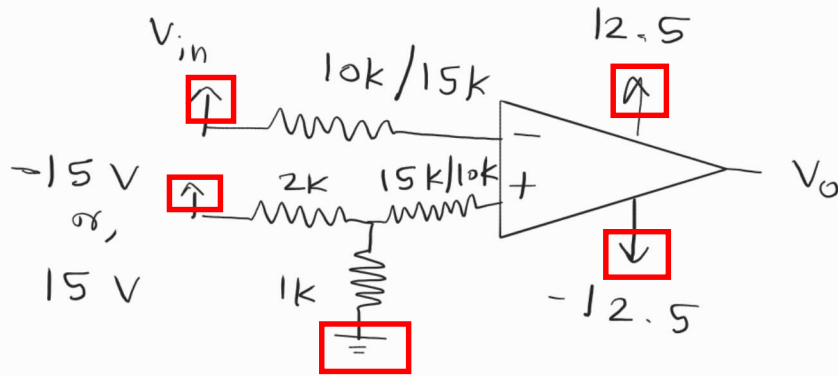


Q1
(a)



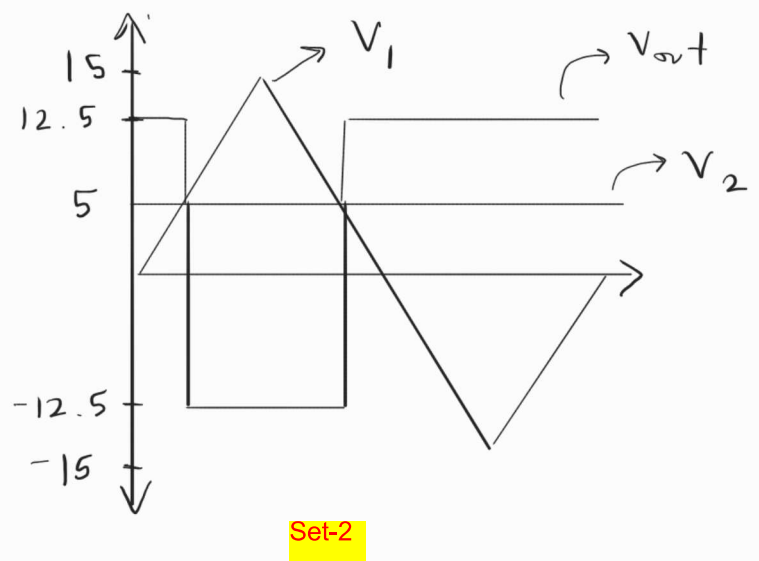
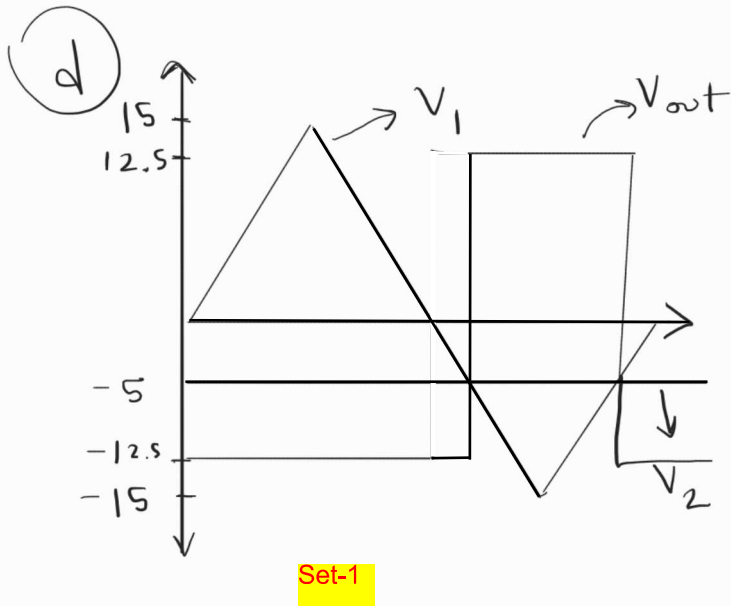
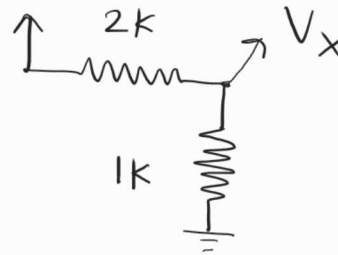
(b) $i_+ = i_- = 0$

(c) Op amp inputs zero current, can be treated independently:

$$V_x = \frac{1}{1+2} \times (15 \text{ or } -15)$$

$$= -5V \text{ or } 5V$$

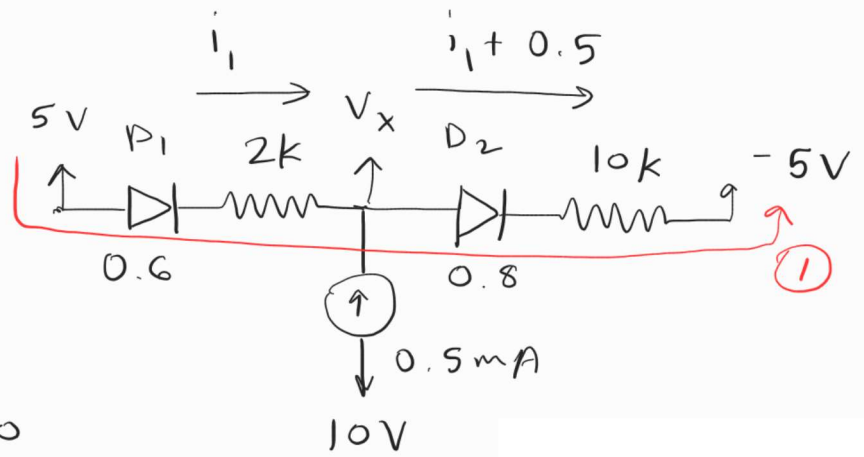
Set-1 Set-2



Q2
Set a:

(a)

D_1, D_2 ON :



Line ①, $12i_1 + 1.4 + 5 \times 10^{-3} = 5 + 5$

$$\Rightarrow 12i_1 = \frac{3.6}{12} = 0.3 \text{ mA}$$

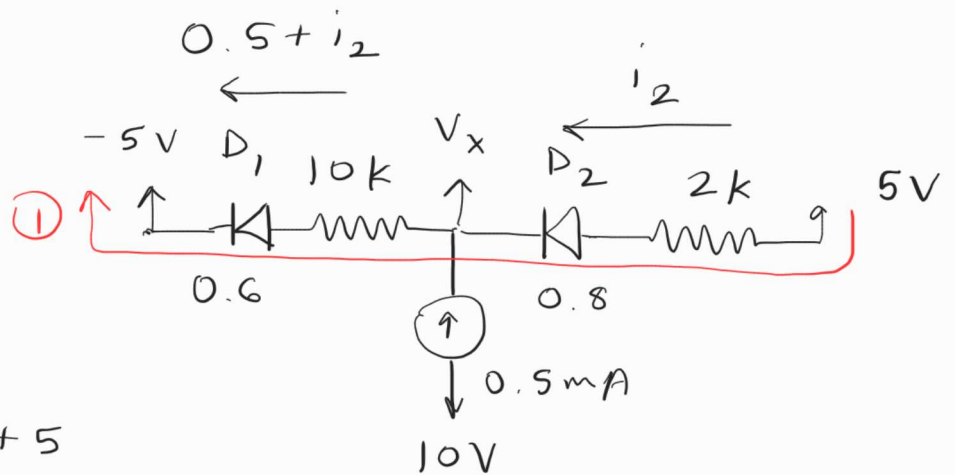
$$V_x = 5 - 0.6 - 2 \times 0.3 = 3.8 \text{ V}$$

$$i_{D_1} = 0.3 \text{ mA}, \quad i_{D_2} = 0.8 \text{ mA}$$

✓✓

Set b:

D_1, D_2 ON :



Line ①,

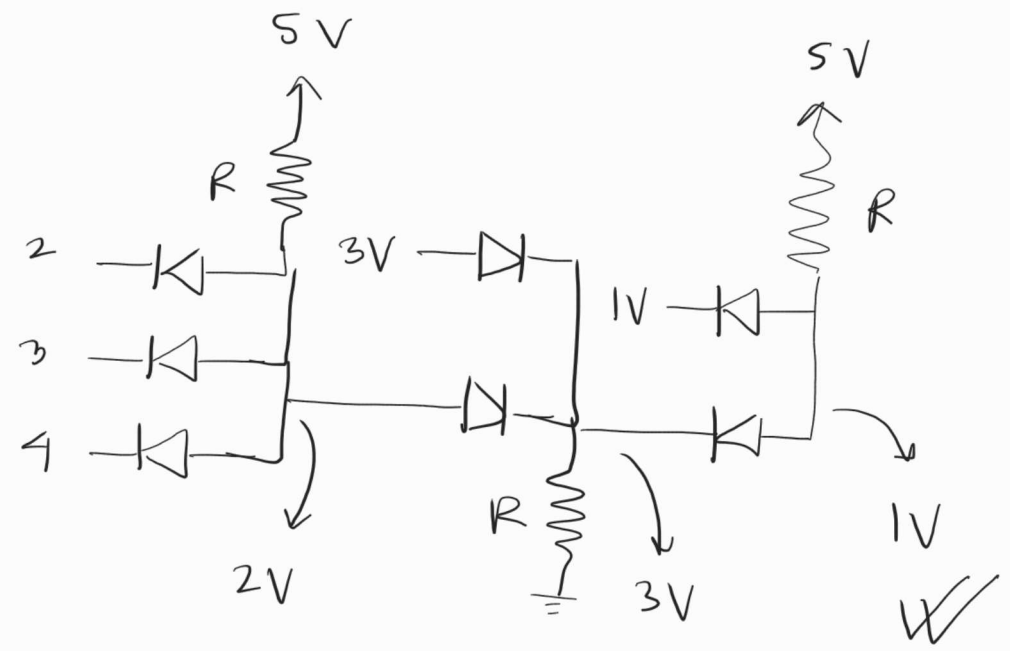
$$12i_2 + 1.4 + 5 = 5 + 5$$

$$\Rightarrow i_2 = \frac{3.6}{12} = 0.3 \text{ mA}$$

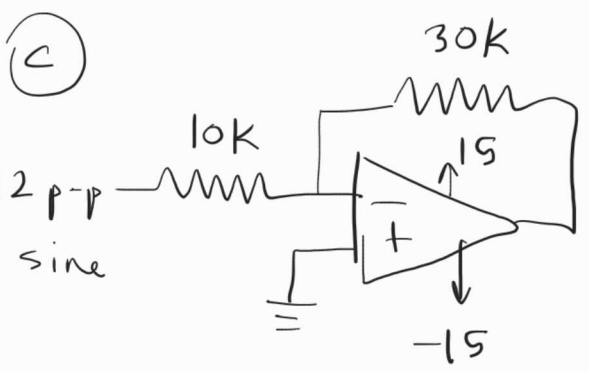
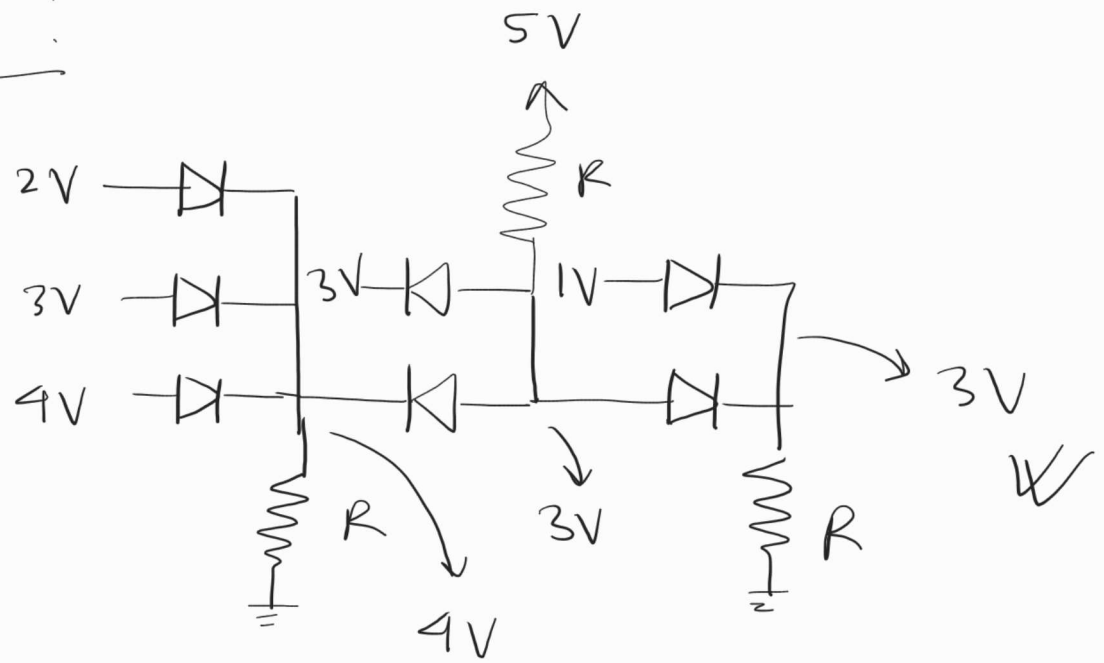
$$i_{D_1} = 0.8 \text{ mA}, \quad i_{D_2} = 0.3 \text{ mA}$$

✓✓

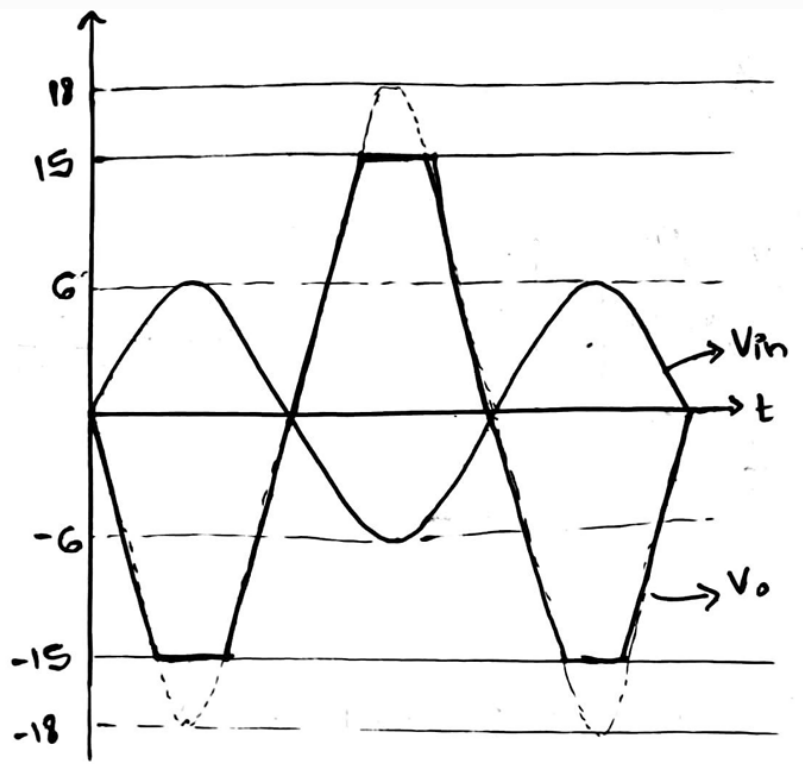
(b) Set A :



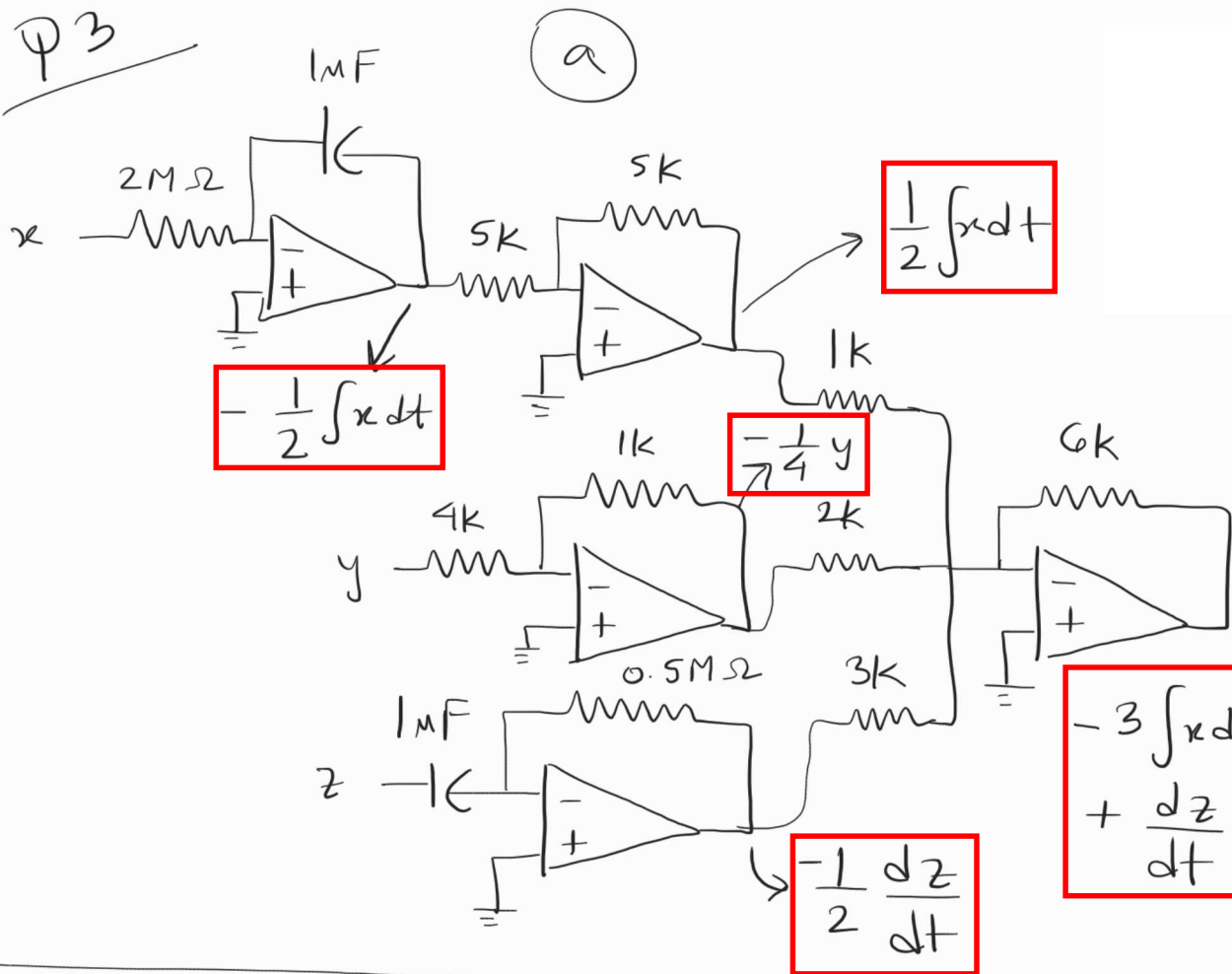
Set B :



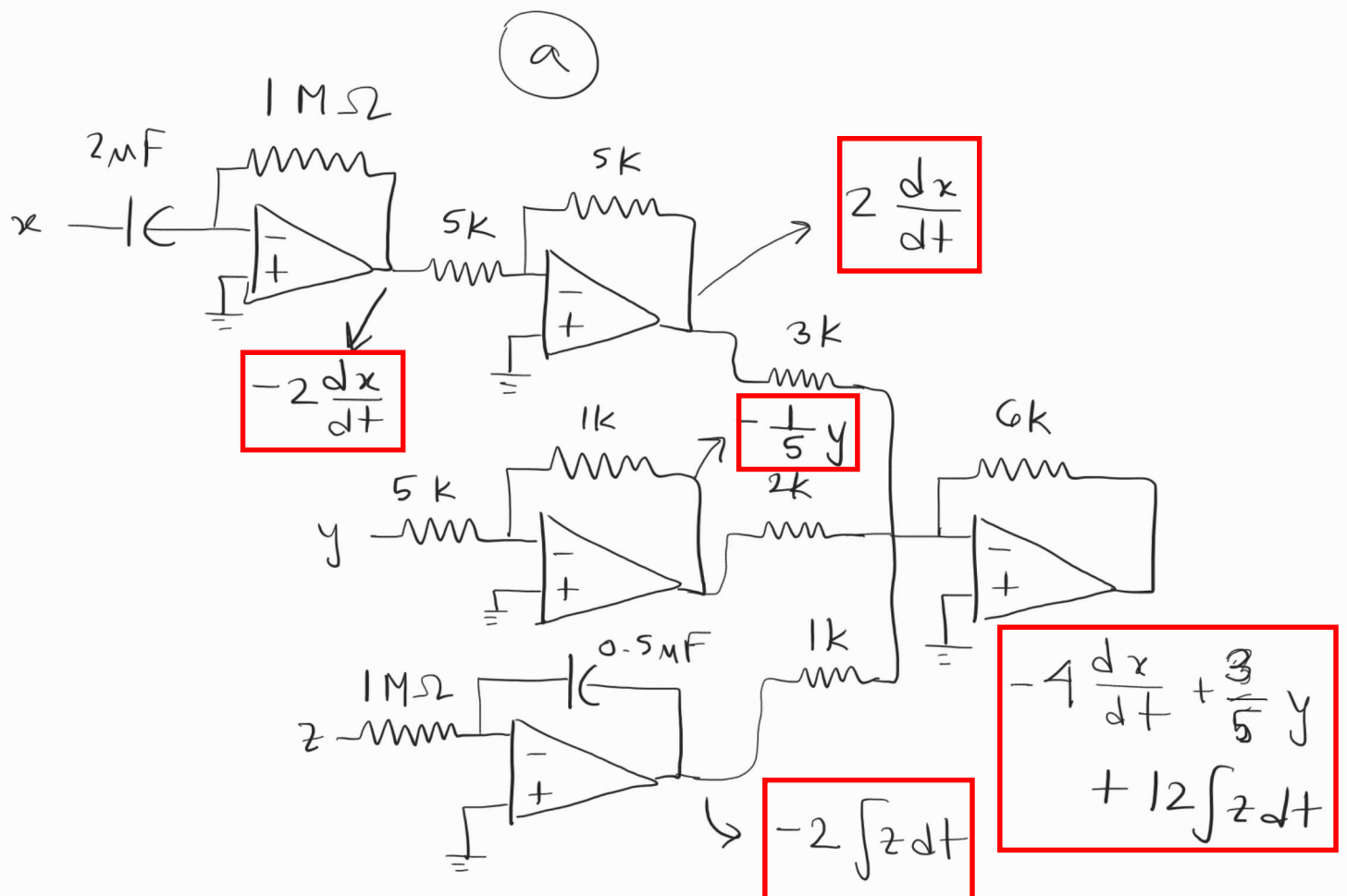
$$V_o = -3V_{in}$$



Q3



(b)



(b)

(A)

If D_1, D_2 ON:

$$V_o = -4V,$$

impossible for D_2 ON.

Then, D_1 ON, D_2 OFF:

$$\textcircled{1}, 12i_1 + 0.7 = 3 + 4$$

$$\Rightarrow i_1 = \frac{6.3}{12} \approx 0.525 \text{ mA} > 0 \quad \checkmark \quad D_1 \text{ ON}$$

$$V_x = 3 - 12 \times i_1 = -3.3V$$

$$V_{D_2} = -3.3 - (-3) = -0.3 < 0.7 \quad \textcircled{B} \quad D_2 \text{ OFF}$$

If D_1, D_2 ON:

$$V_o = -4V,$$

impossible for D_2 ON.

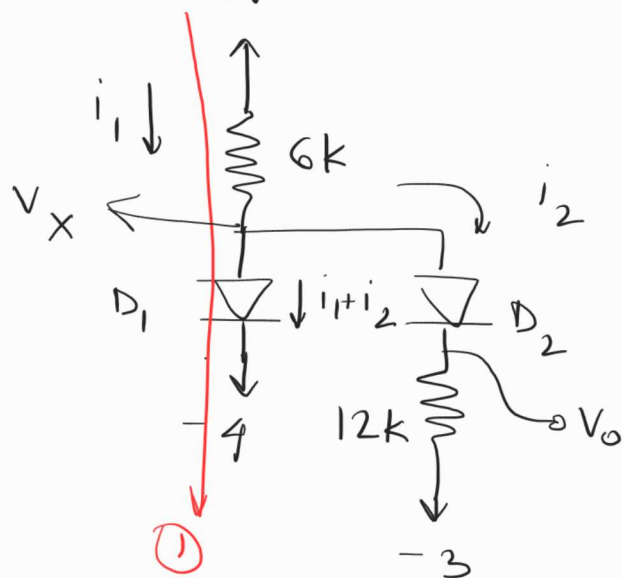
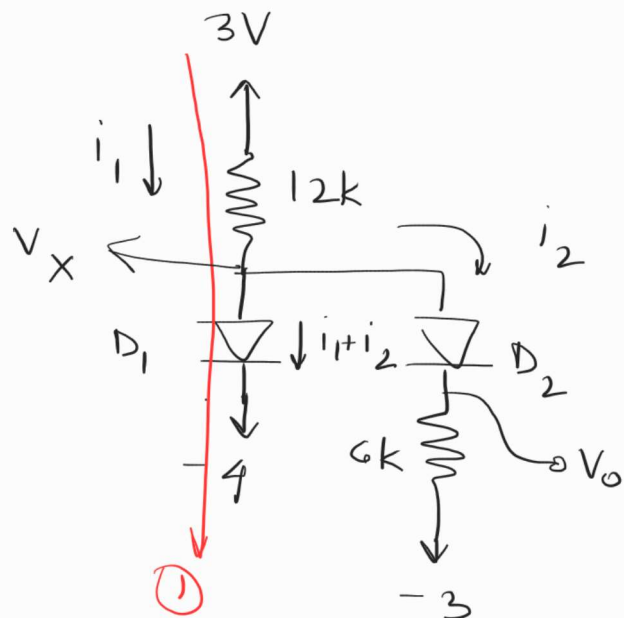
Then, D_1 ON, D_2 OFF:

$$\textcircled{1}, 6i_1 + 0.7 = 3 + 4$$

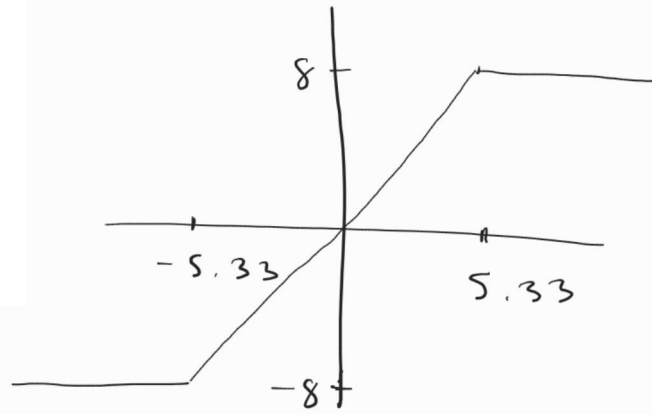
$$\Rightarrow i_1 = \frac{6.3}{6} \approx 1.05 \text{ mA} > 0 \quad \checkmark \quad D_1 \text{ ON}$$

$$V_x = 3 - 6 \times i_1 = -3.3V$$

$$V_{D_2} = -3.3 - (-3) = -0.3 < 0.7 \quad \checkmark \quad D_2 \text{ OFF}$$



c



$$V_o = 1.5 V_{in}$$

$$V_o = 8 \quad V_{in} = 5.33 V$$

$$V_o = -8 \quad V_{in} = -5.33 V$$

Q-4

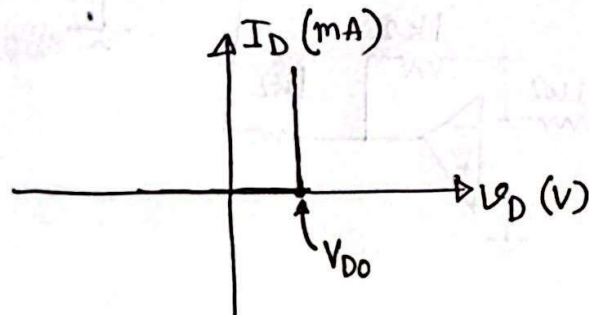
Set-01

(a) $V_o = A(V_+ - V_-) = AV_d$

(b) Linear amplifier is an amplifier whose output is proportional to its input.

Reason for saturation $\rightarrow$ power supply to the amplifier.

(c)



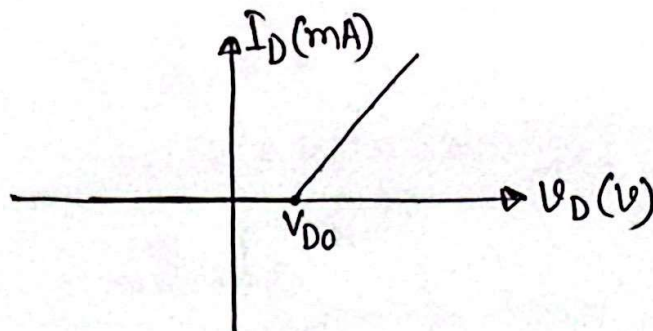
Set-02

4.

(a) Same as set-01

(b) Same as set-01

(c)



Set-01

5(a) $V_{in} = 10\cos(t)$

$V_o = 5\sin(t)$ [By inspection]

gain $= 5/10 = 0.5$

Differentiating $\cos(t)$ gives $-\sin(t)$;

Differentiator gain, $R_f C_1 = 0.5$

$C_1 = \frac{1}{2R_f} \rightarrow \text{Let } R_f = 0.5$

$\therefore C_1 = 1$

} set complementary units
($M\Omega \times \mu F$, $k\Omega \times mF$...)

(Double check -

$$V_o = -0.5 M\Omega \times 1\mu F \times \frac{d}{dt} 10\cos(t)$$

$$= -\frac{10}{2}(-\sin(t))$$

$$= +5\sin(t)$$

(b) Same as 'set-02' 5(b)

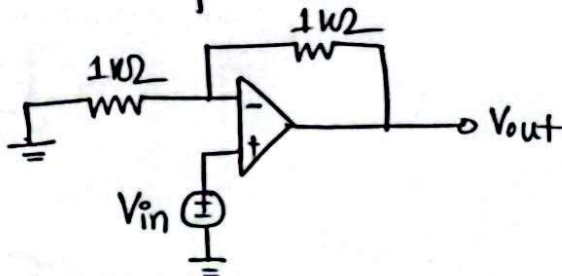
6. (a) Non-inverting amplifier-

$$|\text{Slope}| = \left| \frac{\Delta V_{out}}{\Delta V_{in}} \right| = \left| \frac{10}{5} \right| = 2$$

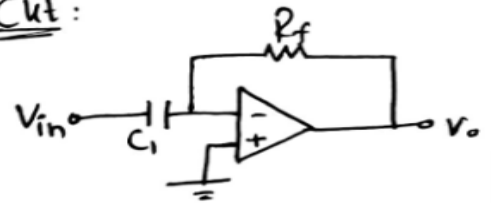
$$\text{Slope} = 2 = \left(1 + \frac{R_f}{R_i} \right) = \text{gain}$$

$$\therefore \frac{R_f}{R_i} = 1$$

$$\therefore R_f = R_i = 1k\Omega \text{ [let's assume]}$$



Ckt:

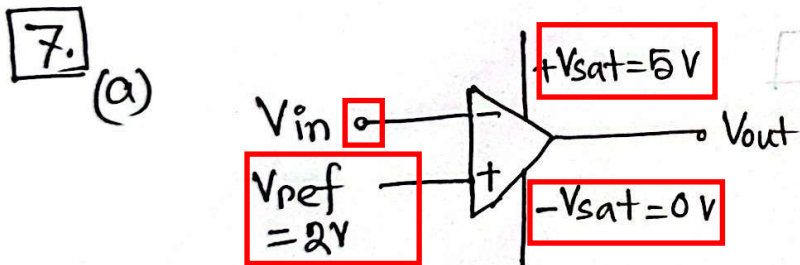
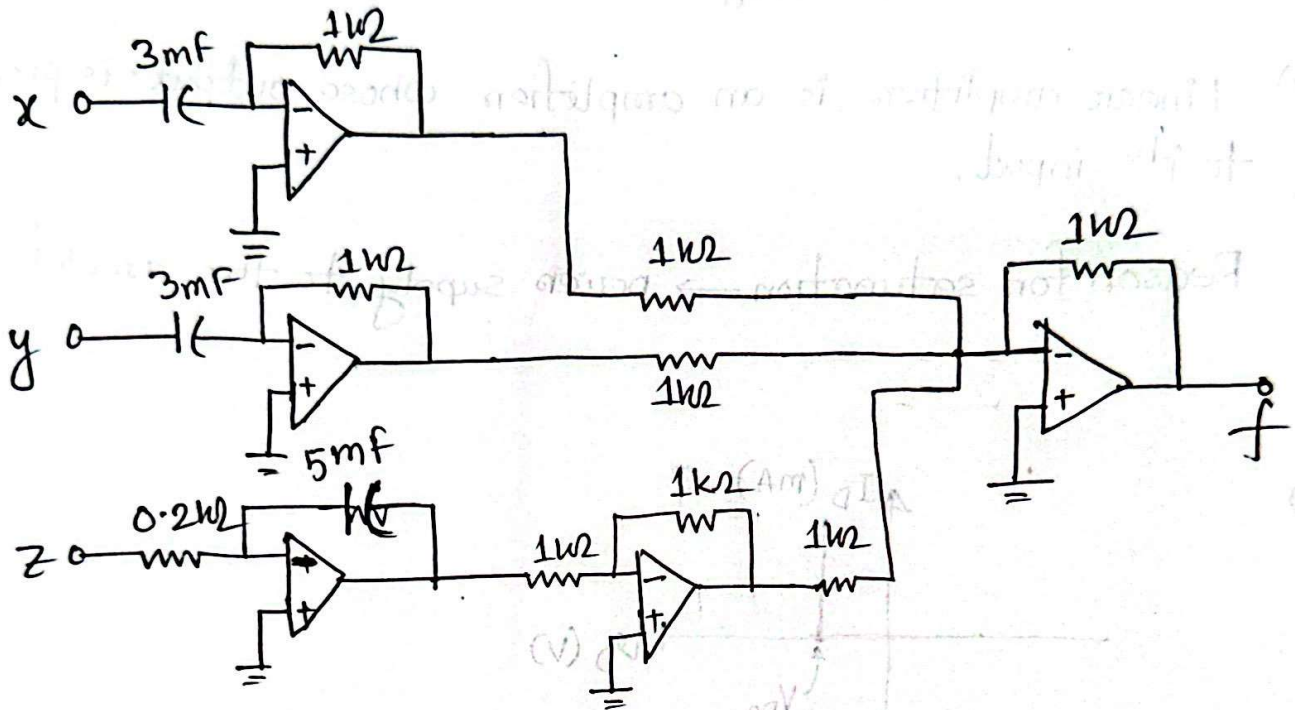


where, $R_f = 0.5 M\Omega$
 $C_1 = 1 \mu F$

(b)

$$f = 3 \frac{d}{dt} (x+y) - 5 \int z dt$$

$$= 3 \frac{d}{dt} x + 3 \frac{d}{dt} y - 5 \int z dt$$



(b) Not possible.

Set B

5. (a) Given,

$$V_{\text{input}} = 5 \sin(t)$$

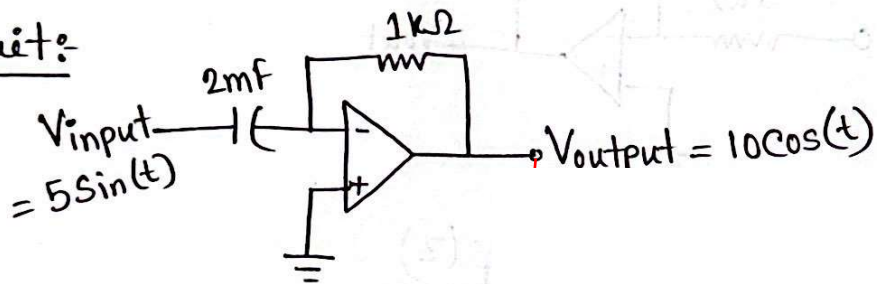
$$V_{\text{output}} = 10 \cos(t)$$

$$\begin{aligned}
 & \xrightarrow{5 \sin(t)} \boxed{\text{Differentiator}} \xrightarrow{10 \cos(t)} \\
 & = 2 \times 5 \frac{d}{dt} \sin(t) \\
 & = RC \times \frac{d}{dt} [5 \sin(t)]
 \end{aligned}$$

$$\therefore RC = 2$$

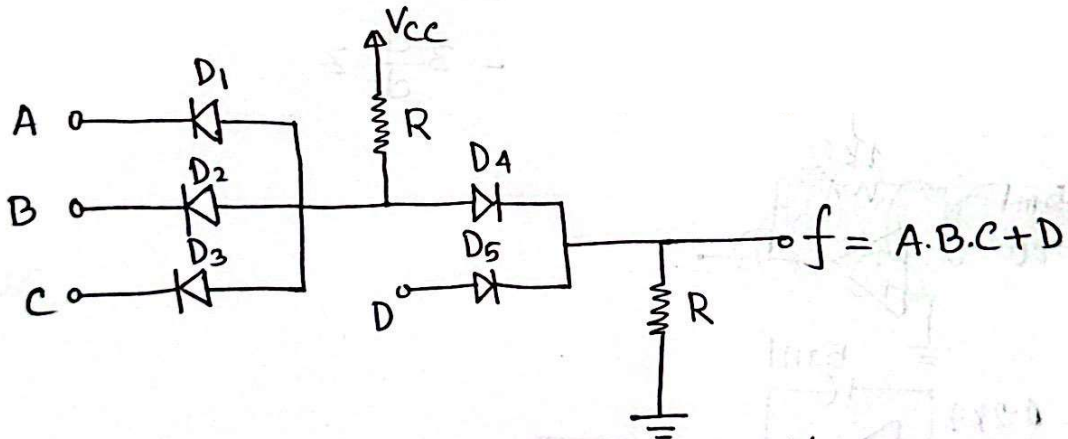
Let's assume, $R = 1 \text{ k}\Omega$; $C = 2 \text{ mF}$

Circuit:-

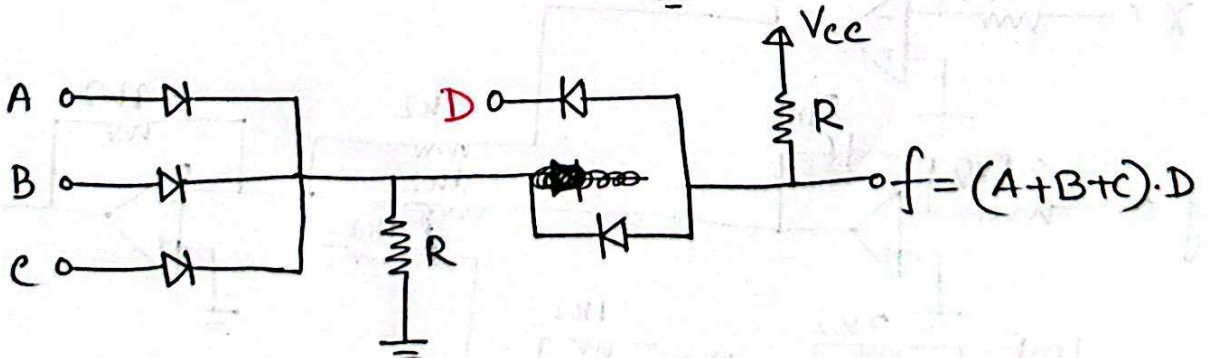


(b)

(i)



(iii)



6.

(a) Inverting Amplifier-

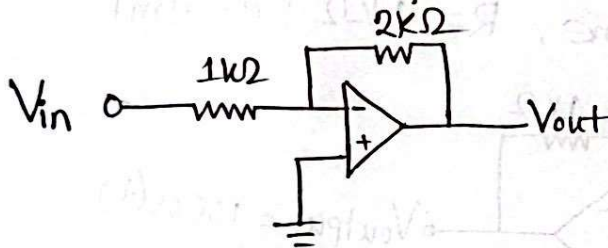
$$|\text{slope}| = \left| \frac{\Delta V_{out}}{\Delta V_{in}} \right| = \left| \frac{10}{5} \right| = 2$$

$$|\text{slope}| = 2 = \frac{R_f}{R_i} = \text{gain}$$

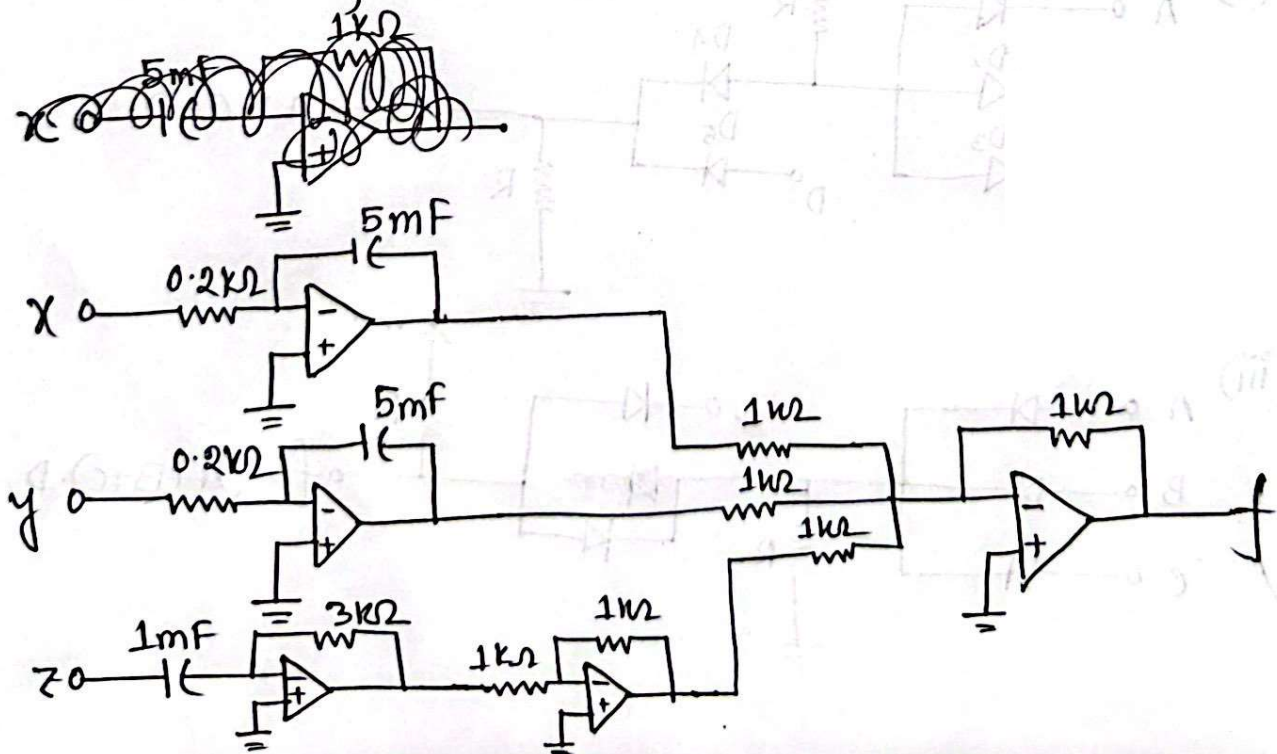
$$\therefore R_f = 2R_i$$

Let's assume, $R_i = 1\text{ k}\Omega$

$$\therefore R_f = 2\text{ k}\Omega$$

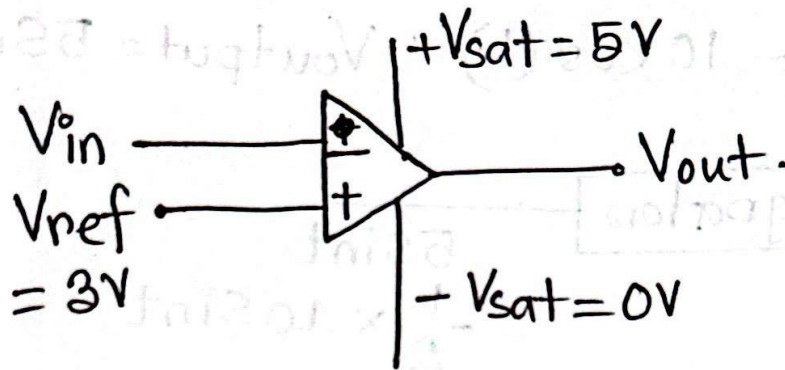


$$\begin{aligned} \text{(b)} \quad f &= 5 \int (x+y) dt - 3 \frac{d}{dt} (z) \\ &= 5 \int x dt + 5 \int y dt - 3 \frac{d}{dt} z \end{aligned}$$



7.

(a)



(b) Not possible.